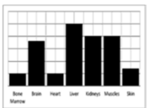
1.2.b. Preparation and training methods in relation to improving and maintaining physical activity and performance

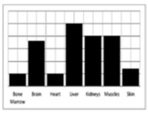
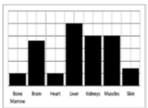
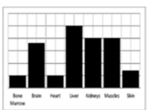
Learners will develop their knowledge and understanding of aerobic training, methods of evaluating aerobic capacity and factors affecting VO_2max, as well as applying the importance of this training to physical activities and sports.

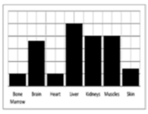
Strength and flexibility training will also be covered, including knowledge and understanding of the types of strength and flexibility training, factors that affect strength and flexibility and methods of evaluating strength and flexibility. Learners will also be able to understand how training can be used to develop strength and flexibility through different training activities and how the body adapts to such training.

Learners will also develop their knowledge and understanding of the periodisation of training and how to plan personal health and fitness programmes.

Learners will also develop their knowledge and understanding of the impact of training on lifestyle-related diseases that affect the cardiovascular and respiratory systems.

Topic Area	**Content**
Aerobic training  	• aerobic capacity and maximal oxygen uptake (VO_2max) • how VO_2max is affected by: ○ individual physiological make-up ○ training ○ age ○ gender • methods of evaluating aerobic capacity: ○ laboratory test of VO_2max using direct gas analysis ○ NCF multi-stage fitness test ○ Queen's College step test ○ Cooper 12 minute run • intensity and duration of training used to develop aerobic capacity: ○ continuous training ○ high intensity interval training (HIIT) • the use of target heart rates as an intensity guide

Topic Area	Content
Aerobic training cont.  	• physiological adaptations from aerobic training: ○ cardiovascular ○ respiratory ○ muscular ○ metabolic • activities and sports in which aerobic capacity is a key fitness component.
Strength training  	• types of strength: ○ strength endurance ○ maximum strength ○ explosive/elastic strength ○ static and dynamic strength • factors that affect strength: ○ fibre type ○ cross sectional area of the muscle • methods of evaluating each type of strength: ○ grip strength dynamometer ○ 1 Repetition Maximum(1RM) ○ press up or sit-up test ○ vertical jump test • training to develop strength: ○ repetitions ○ sets ○ resistance guidelines used to improve each type of strength ○ use of multi-gym ○ weights ○ plyometrics ○ circuit/interval training: – work intensity – work duration – relief interval – number of work/relief intervals • physiological adaptations from strength training: ○ muscle and connective tissues ○ neural ○ metabolic • activities and sports in which strength is a key fitness component.
Flexibility training  	• types of flexibility: ○ static flexibility (active and passive) ○ dynamic flexibility • factors that affect flexibility: ○ type of joint ○ length of surrounding connective tissue ○ age ○ gender

Topic Area	Content
Flexibility training cont.  	• methods of evaluating flexibility: ○ sit and reach test ○ goniometer • training used to develop flexibility: ○ passive stretching ○ proprioceptive neuromuscular facilitation (PNF) ○ static stretching ○ dynamic stretching ○ ballistic stretching ○ isometric stretching • physiological adaptations from flexibility training: ○ muscle and connective tissues • activities and sports in which flexibility is a key fitness component.
Periodisation of training 	• periodisation cycles: ○ macrocycle ○ mesocycle ○ microcycle • phases of training: ○ preparatory ○ competitive ○ transition • tapering to optimise performance • how to plan personal health and fitness programmes for aerobic, strength and flexibility training.
Impact of training on lifestyle diseases 	• the effect of training on lifestyle diseases: ○ cardiovascular system : – coronary heart disease (CHD) – stroke – atherosclerosis – heart attack ○ respiratory system – asthma – chronic obstructive pulmonary disease (COPD).